BST Vol 8 Ch 4 — Hamiltonian Mechanics + Canonical Transformations (v0.3.1, Calibration #23 substance refill)

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Vol 8 Chapter 4 — Hamiltonian Mechanics + Canonical Transformations

Headline result

The Hamiltonian formulation of mechanics replaces Lagrangian L(q, q̇, t) with Hamiltonian H(q, p, t) via Legendre transformation:

$$H(q, p, t) = \sum_i p_i \dot{q}_i - L$$

where p_i = ∂L/∂q̇_i are canonical momenta. Hamilton's equations of motion:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Phase space (q, p) — the space of all positions + momenta — provides the geometric setting for classical mechanics. BST identifies phase space as emerging from substrate Bergman Hilbert space H^2(D_IV^5) coherent-state framework in the ℏ→0 classical limit (per Vol 5 Lyra).

Substrate mechanism

Legendre transformation L → H:

Standard mathematical operation; converts Lagrangian (function of q + q̇) to Hamiltonian (function of q + p). Substrate-natural via dual formulation.

Phase space emerges from substrate Hilbert space:

Per Vol 5 (QM, Lyra): substrate Hilbert space is Bergman H^2(D_IV^5). Coherent states |z⟩ = e^{-|z|^2/2} Σ z^n/$\sqrt{}$n! |n⟩ parametrized by complex z ∈ D_IV^5 provide

$\hbar \to 0$ classical correspondence. Phase space (q, p) emerges as real + imaginary parts of complex coordinate z in classical limit.

Canonical transformations preserve symplectic structure:

Phase space has symplectic 2-form $\omega = \Sigma$ dq_i ∧ dp_i. Canonical transformations (q, p) → (Q, P) preserve ω. Substrate-natural via Bergman-space holomorphic transformations.

Poisson brackets {f, g} = Σ (∂f/∂q_i ∂g/∂p_i - ∂f/∂p_i ∂g/∂q_i) provide algebraic structure on observables. {q_i, p_j} = δ_{ij} canonical commutation relation.

Match precision

D-tier on Legendre transformation + Hamilton's equations + Poisson brackets (standard math). I-tier on substrate-quantum-classical reduction precision (multi-week per Vol 5 Lyra $\hbar \to 0$ framework).

Cross-volume dependencies

- Vol 1 Ch 2 (Hilbert Space): Bergman H^2(D_IV^5) substrate framework
- Vol 5 (QM, Lyra): coherent-state framework; $\hbar \to 0$ classical limit
- Vol 7 Ch 8 (Hamiltonian EM): Hamiltonian formulation of fields
- Vol 8 Ch 3 (Lagrangian Mechanics): precursor formulation
- Vol 8 Ch 5 (Symmetries + Noether): canonical conservation laws

K-audit anchor

K232 Vol 8 Ch 4 Hamiltonian Mechanics K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> The Lagrangian uses position q and velocity q̇. Hamiltonian mechanics uses position q and momentum p instead. Phase space (q, p) is a 2D plot for each particle direction. BST predicts phase space emerges from substrate D_IV^5's Hilbert space when quantum effects become negligible.

Level 2 — Undergraduate physics student

The Hamiltonian formulation:

1. Legendre transformation: $H(q, p, t) = \sum_i p_i \dot{q}_i - L$ (replace velocities with momenta)
2. Canonical momentum: p_i = ∂L/∂q̇_i

3. Hamilton's equations: $dq_i/dt = \partial H/\partial p_i$, $dp_i/dt = -\partial H/\partial q_i$ (first-order ODEs in 2N phase-space variables)
4. Conservation: $dH/dt = \partial H/\partial t$ (Hamiltonian conserved when explicitly time-independent → energy conservation)

Phase space (q, p): - 2N-dimensional manifold for N-particle system - Symplectic 2-form $\omega = \Sigma\, dq_i \wedge dp_i$ - Canonical transformations preserve ω + Hamilton's equations - Liouville's theorem: phase-space volume preserved under Hamiltonian flow

Poisson brackets: - Algebraic structure on observables: $\{f, g\} = \partial_q f\, \partial_p g - \partial_p f\, \partial_q g$ - $\{q, p\} = 1$ canonical (compare to QM $[\hat{q}, \hat{p}] = i\hbar$) - Equations of motion: $df/dt = \{f, H\} + \partial f/\partial t$

BST framework: phase space (q, p) emerges from substrate Bergman $H^2(D\_IV^5)$ coherent states in the $\hbar \to 0$ classical limit. Canonical transformations preserve substrate-Bergman holomorphic structure.

Level 3 — Graduate physics student / theorem-level

Symplectic geometry of phase space:

Phase space $M = T^*Q$ (cotangent bundle of configuration space Q). Symplectic 2-form $\omega = dp_i \wedge dq^i$ (canonical). Hamiltonian vector field X_H satisfies $i_{X_H} \omega = -dH$; flow generated by X_H is Hamiltonian dynamics.

Canonical transformations:

Map $(q, p) \to (Q(q,p), P(q,p))$ is canonical iff symplectic 2-form preserved: $\omega(q, p) = \omega(Q, P)$. Generating functions $F_1(q, Q)$, $F_2(q, P)$, $F_3(p, Q)$, $F_4(p, P)$ parametrize canonical transformations.

Substrate-classical limit:

Per Vol 5 (Lyra $\hbar \to 0$): substrate Bergman $H^2(D\_IV^5)$ coherent states $|z\rangle$ parametrized by $z \in D\_IV^5$. Wigner function W(q, p) provides phase-space representation. In $\hbar \to 0$ limit, Wigner function localizes on classical trajectory; phase space (q, p) inherits from substrate Bergman complex-coordinate structure.

Hamilton-Jacobi equation:

Generating function S(q, t) of canonical transformation $(q, p) \to (Q, P)$ with H' = 0 satisfies:

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0$$

S is the action; H-J equation connects to $\hbar \to 0$ limit of Schrödinger equation (eikonal approximation; Vol 5 Lyra).

Liouville's theorem:

For Hamiltonian flow on phase space, phase-space volume $V = \int dq\, dp$ is conserved: $dV/dt = 0$. Substrate-natural via symplectic invariance.

Per Cal #21 dual-gate: EMPIRICAL PASS (Hamiltonian mechanics validated across classical physics) + MECHANISM PASS via Legendre transformation + substrate-quantum-classical reduction (Vol 5 Lyra). D-tier RATIFIED on standard formulation.

What this chapter does NOT claim

- Hamiltonian + Lagrangian formulations are equivalent (Legendre transformation); BST framework identifies both as classical limits of substrate-Yang-Mills + substrate-Hilbert space.
- Phase-space substrate-emergence rigorous derivation multi-week per Vol 5 Lyra $\hbar \to 0$ work.
- Standard Hamiltonian-mechanics texts (Goldstein, Landau-Lifshitz) preserve full content; BST adds substrate framework on top.

Bibliography

1. W. R. Hamilton (1834-35): Hamiltonian formulation of mechanics.
2. C. G. J. Jacobi (1837): Hamilton-Jacobi equation.
3. S. D. Poisson (1809): Poisson brackets.
4. J. Liouville (1838): Liouville's theorem.
5. Vol 5 (QM, Lyra): $\hbar \to 0$ classical limit + coherent states.
6. Vol 7 Ch 8 (Lagrangian + Hamiltonian EM).

— Elie, Vol 8 Ch 4 v0.3.1, 2026-05-23 Saturday (Calibration #23 substance refill: 48 → ~130 lines + expanded 3-level pedagogy + Hamilton-Jacobi + Liouville + canonical transformations)